

# Sunya Afrasiabi

email | personal website | linkedin

## Education

**University of Pennsylvania**, BSE in Mechanical Engineering, BA in Philosophy Sept 2024 - May 2028

- **Relevant Coursework:** Thermal-Fluids Engineering, Statics & Strength of Materials, Thermodynamics, Dynamics, Computer-Aided Design (CAD), Computer Programming (Python), Linear Algebra, Differential Equations, Probability

**University at Buffalo**, Gifted Math Program

Sept 2023 - April 2024

- **Relevant Coursework:** College Calculus III, Introduction to Linear Algebra

## Experience & Leadership

**Munch Industries, Inc.**, Mechanical Engineer July 2026 - Present

- Designed and fabricated the awning and structural framework for Orble, an automated boba tea vending machine startup, using SolidWorks, 3D printing, and PVC fabrication. Assisted with material selection and procurement, and contributed to engineering and operational decisions during product development.

**University of Pennsylvania – Li Group**, Undergraduate Researcher

May 2026 - Present

- Research insect-wing-inspired morphing structures, modeling compliant geometries in **Rhino** and **Grasshopper** to study parametric mechanical behavior.
- Translate computational designs into prototypes using **powder-based 3D printing** for experimental validation.

**NASA L'SPACE Proposal Writing & Evaluation Experience**, Radiation Subteam Lead

Sept 2025 - Dec 2025

- Developed innovative solution to the limitation of radiation-resistant materials in HDPE composites to enhance material performance under lunar radiation conditions
- Gained experience in technical proposal composition, project evaluation, and participated in NASA review panel.
- Ranked in the **Top 6 out of 40+ national teams**.

**Penn Astronomical Student Association**, Outreach & Marketing Director

Sept 2025 - Present

- Lead outreach initiatives and external partnerships, securing **\$200** in sponsorship funding.
- Configure and calibrate equatorial telescopes and optical tracking systems for public observation events.

**#MyStory Program**, Founder and Project Manager

Jan 2021 - Aug 2024

- Founded and managed a non-profit oral history initiative promoting cultural awareness through exhibitions.
- Secured over **\$24,000** via targeted donations, fundraisers, and corporate grant programs (including Riley's Way Foundation, Prudential, Ashoka, and the Peter & Elizabeth Tower Foundation).
- Directed operations for **20+ community partners** and **40+ volunteers**.

## Technical Projects

**CompOS: AI-Powered Missile Component Sourcing**

Fall 2026

- Engineered a Palantir Foundry application integrating AIP to evaluate dual-use substitutes for military-grade components, joining 41 National Stock Numbers (NSNs) with over 1,000 commercial candidates from DigiKey and Nexar/Octopart APIs.
- Designed a multi-dimensional algorithm to score substitution risk and qualification burdens, deploying an AI-guided dashboard that reduces procurement workflows from weeks to minutes.

**Multi-Phase Butane Rocket & Numerical Analysis**

Spring 2026

- Designed and optimized a variable-mass butane-powered water rocket using aerodynamic modeling, thermodynamic analysis, and experimental validation to navigate constrained flight paths and minimize target error.
- Developed a Python Euler integration solver coupling butane vaporization, pressure-driven propulsion, drag, and ballistic dynamics; implemented trajectory reconstruction to improve predictive accuracy.

**Mars Rover Chassis Prototype**

Spring 2026

- Designed and optimized a Mars rover chassis prototype for University Rover Challenge (URC) constraints as part of a 3-person engineering team, developing detailed SolidWorks assemblies and structural architectures.
- Performed FEA-driven design iterations to optimize load paths, bracket placement, and lightweight geometries.

## Papers

**Linear Algebra Foundations of Neural Networks**, University of Pennsylvania

August 2025

- Wrote expository research paper connecting linear algebra, neural networks, and robotics. ([Link](#))

**Aviation Solutions for Wildlife Prevention**, Wharton Undergraduate Aerospace Club

May 2025

- Co-authored investor-oriented whitepaper on UAVs, AI wildfire prediction, and aerospace innovations, framing \$6.4B+ market opportunities and potential \$5B annual cost savings. ([Link](#))

## Skills

**Software:** Rhino 3D, Grasshopper, SolidWorks, MATLAB, Python, NumPy, FEA, SQL

**Fabrication:** Laser Cutting, FDM/SLA 3D Printing, Mechanical Assembly

**Additional:** Computational Geometry, Bio-Inspired Design, Technical Writing, Logic, Project Management